

Computational Companion to “Generalized Quasi-Symmetric Nets: A Constructive Approach to the Equimodular Elliptic Type of $n \times m$ Kokotsakis Polyhedra” — Helper

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Tested on: Mathematica 14.0

Section 1

Results of Section 1 are used in the proof of Lemmas 1-2
(see paper)

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In[*]:= (* =====*)
(*PART 1:INITIALIZATION& DOMAIN ASSUMPTIONS*)
(* =====*)
ClearAll[ $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$ ,  $\sigma$ , a, b, c, d, r, s, f,
  M, assumptions, isTrueQ, statusChip, row, identities];
 $\sigma = (\alpha + \beta + \gamma + \delta) / 2$ ;
a = Sin[ $\alpha$ ] / Sin[ $\sigma - \alpha$ ]; b = Sin[ $\beta$ ] / Sin[ $\sigma - \beta$ ];
c = Sin[ $\gamma$ ] / Sin[ $\sigma - \gamma$ ]; d = Sin[ $\delta$ ] / Sin[ $\sigma - \delta$ ];
r = a d; s = c d; f = a c; M = a b c d;
(*Domain assumptions:the flat angles lie in (0,Pi)*)
assumptions = {0 <  $\alpha$  < Pi, 0 <  $\beta$  < Pi, 0 <  $\gamma$  < Pi, 0 <  $\delta$  < Pi};
isTrueQ[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions  $\rightarrow$  assumptions]];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background  $\rightarrow$  RGBColor[0.13, 0.55, 0.13], FrameStyle  $\rightarrow$  None,
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RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
(* =====*)
(*PART 2:THE SIX RATIO IDENTITIES*)
(* =====*)
identities = {"Identity 1", HoldForm[Sin[α] Sin[δ] / (Sin[β] Sin[γ])],
  HoldForm[r (r - M) / (M (r - 1))], isTrueQ},
{"Identity 2", HoldForm[Sin[σ - α] Sin[σ - δ] / (Sin[σ - β] Sin[σ - γ])],
  HoldForm[(r - M) / (r (r - 1))], isTrueQ},
{"Identity 3", HoldForm[Sin[γ] Sin[δ] / (Sin[α] Sin[β])],
  HoldForm[s (s - M) / (M (s - 1))], isTrueQ},
{"Identity 4", HoldForm[Sin[σ - γ] Sin[σ - δ] / (Sin[σ - α] Sin[σ - β])],
  HoldForm[(s - M) / (s (s - 1))], isTrueQ},
{"Identity 5", HoldForm[Sin[α] Sin[γ] / (Sin[β] Sin[δ])],
  HoldForm[f (f - M) / (M (f - 1))], isTrueQ},
{"Identity 6", HoldForm[Sin[σ - α] Sin[σ - γ] / (Sin[σ - β] Sin[σ - δ])],
  HoldForm[(f - M) / (f (f - 1))], isTrueQ}};
(* =====*)
(*PART 3:VERIFICATION TABLE*)
(* =====*)
row[{desc_, lhs_, rhs_, test_}] :=
  {Style[desc, 13], Item[Style[TraditionalForm[lhs], 16], Alignment → Center],
  Item[Style[TraditionalForm[rhs], 16], Alignment → Center],
  statusChip[test[ReleaseHold@lhs, ReleaseHold@rhs]]};
Grid[
  Join[{{Style["Description", Bold, 14], Style["Left Hand Side (LHS)", Bold, 14],
    Style["Right Hand Side (RHS)", Bold, 14],
    Style["Equality Status", Bold, 14]}}, row /@ identities],
  Frame → All, FrameStyle → GrayLevel[0.75],
  Background → {None, {1 → RGBColor[0.99, 0.96, 0.88]}},
  Spacings → {2.2, 2.8},
  Alignment → {{Left, Center, Center, Center}, Center},
  ItemStyle → Directive[FontFamily → "Helvetica", 13]]

```

Out[]=

Description	Left Hand Side (LHS)	Right Hand Side (RHS)	Equality Status
Identity 1	$\frac{\sin(\alpha) \sin(\delta)}{\sin(\beta) \sin(\gamma)}$	$\frac{r(r-M)}{M(r-1)}$	Verified
Identity 2	$\frac{\sin(\sigma-\alpha) \sin(\sigma-\delta)}{\sin(\sigma-\beta) \sin(\sigma-\gamma)}$	$\frac{r-M}{r(r-1)}$	Verified
Identity 3	$\frac{\sin(\gamma) \sin(\delta)}{\sin(\alpha) \sin(\beta)}$	$\frac{s(s-M)}{M(s-1)}$	Verified
Identity 4	$\frac{\sin(\sigma-\gamma) \sin(\sigma-\delta)}{\sin(\sigma-\alpha) \sin(\sigma-\beta)}$	$\frac{s-M}{s(s-1)}$	Verified
Identity 5	$\frac{\sin(\alpha) \sin(\gamma)}{\sin(\beta) \sin(\delta)}$	$\frac{f(f-M)}{M(f-1)}$	Verified
Identity 6	$\frac{\sin(\sigma-\alpha) \sin(\sigma-\gamma)}{\sin(\sigma-\beta) \sin(\sigma-\delta)}$	$\frac{f-M}{f(f-1)}$	Verified

Section 2

Results of Section 2 are used in the proof of Lemma 2 (see paper)

```

In[ ]:= (* =====*)
(*PART 1:INITIALIZATION*)
(* =====*)
ClearAll[f, l, m, M, isTrueQ, statusChip, row, identities];
(*The identities below are identities of
rational functions in Subscript[f,l],Subscript[f,m],M;
the subscripted symbols remain inert,so no domain assumptions are needed*)
isTrueQ[lhs_, rhs_] := TrueQ[Simplify[lhs == rhs]];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
(* =====*)
(*PART 2:THE FOUR FACTORIZATION IDENTITIES*)

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(* =====*)
identities = {"Factorization 1",
  HoldForm[Subscript[f, l] (Subscript[f, l] - M) / (M (Subscript[f, l] - 1)) -
    Subscript[f, m] (Subscript[f, m] - M) / (M (Subscript[f, m] - 1))],
  HoldForm[(Subscript[f, l] - Subscript[f, m])
    (Subscript[f, l] × Subscript[f, m] - (Subscript[f, l] + Subscript[f, m]) + M) /
    (M (Subscript[f, l] - 1) (Subscript[f, m] - 1))],
  isTrueQ}, {"Factorization 2", HoldForm[
  (Subscript[f, l] (Subscript[f, l] - M) / (M (Subscript[f, l] - 1))) *
    (Subscript[f, m] (Subscript[f, m] - M) / (M (Subscript[f, m] - 1))) - 1],
  HoldForm[(Subscript[f, l] × Subscript[f, m] - M) (Subscript[f, l] ×
    Subscript[f, m] - M (Subscript[f, l] + Subscript[f, m] - 1)) /
    (M^2 (Subscript[f, l] - 1) (Subscript[f, m] - 1))], isTrueQ},
{"Factorization 3", HoldForm[(Subscript[f, l] - M) /
  (Subscript[f, l] (Subscript[f, l] - 1)) -
  (Subscript[f, m] - M) / (Subscript[f, m] (Subscript[f, m] - 1))],
  HoldForm[(Subscript[f, m] - Subscript[f, l]) (Subscript[f, l] ×
    Subscript[f, m] - M (Subscript[f, l] + Subscript[f, m] - 1)) /
    (Subscript[f, l] × Subscript[f, m] (Subscript[f, l] - 1)
    (Subscript[f, m] - 1))], isTrueQ}, {"Factorization 4",
  HoldForm[((Subscript[f, l] - M) / (Subscript[f, l] (Subscript[f, l] - 1))) *
    ((Subscript[f, m] - M) / (Subscript[f, m] (Subscript[f, m] - 1))) - 1],
  HoldForm[(Subscript[f, l] × Subscript[f, m] - M)
    (Subscript[f, l] + Subscript[f, m] - Subscript[f, l] × Subscript[f, m] - M) /
    (Subscript[f, l] × Subscript[f, m] (Subscript[f, l] - 1)
    (Subscript[f, m] - 1))], isTrueQ}}];

(* =====*)
(*PART 3:VERIFICATION TABLE*)
(* =====*)
row[{desc_, lhs_, rhs_, test_}] :=
  {Style[desc, 13], Item[Style[TraditionalForm[lhs], 16], Alignment → Center],
    Item[Style[TraditionalForm[rhs], 16], Alignment → Center],
    statusChip[test[ReleaseHold@lhs, ReleaseHold@rhs]]];
Grid[
  Join[{{Style["Description", Bold, 14], Style["Left Hand Side (LHS)", Bold, 14],
    Style["Right Hand Side (RHS)", Bold, 14],
    Style["Equality Status", Bold, 14]}}, row /@ identities],
  Frame → All, FrameStyle → GrayLevel[0.75],
  Background → {None, {1 → RGBColor[0.99, 0.96, 0.88]}},
  Spacings → {2.2, 2.8},
  Alignment → {{Left, Center, Center, Center}, Center},
  ItemStyle → Directive[FontFamily → "Helvetica", 13]]

```

Out[]:=

Description	Left Hand Side (LHS)	Right Hand Side (RHS)	Equality Status
Factorization 1	$\frac{f_l(f_l-M)}{M(f_l-1)} - \frac{f_m(f_m-M)}{M(f_m-1)}$	$\frac{(f_l-f_m)(f_l f_m - (f_l+f_m)+M)}{M(f_l-1)(f_m-1)}$	Verified
Factorization 2	$\frac{(f_l(f_l-M))(f_m(f_m-M))}{(M(f_l-1))(M(f_m-1))} - 1$	$\frac{(f_l f_m - M)(f_l f_m - M(f_l+f_m-1))}{M^2(f_l-1)(f_m-1)}$	Verified
Factorization 3	$\frac{f_l-M}{f_l(f_l-1)} - \frac{f_m-M}{f_m(f_m-1)}$	$\frac{(f_m-f_l)(f_l f_m - M(f_l+f_m-1))}{f_l f_m(f_l-1)(f_m-1)}$	Verified
Factorization 4	$\frac{(f_l-M)(f_m-M)}{(f_l(f_l-1))(f_m(f_m-1))} - 1$	$\frac{(f_l f_m - M)(f_l+f_m-f_l f_m - M)}{f_l f_m(f_l-1)(f_m-1)}$	Verified

Section 3

Results of Section 3 are used in the proof of Lemma 4 (see paper)

```

In[ ]:= (* =====*)
(*PART 1:INITIALIZATION& DOMAIN ASSUMPTIONS*)
(* =====*)
ClearAll[α, β, γ, δ, σ, ε, a, b, c, d, r, s, f, M, rad, sq,
  assumptions, isTrueQ, isTrueQeps, statusChip, row, identities];
σ = (α + β + γ + δ) / 2;
a = Sin[α] / Sin[σ - α]; b = Sin[β] / Sin[σ - β];
c = Sin[γ] / Sin[σ - γ]; d = Sin[δ] / Sin[σ - δ];
r = a d; s = c d; f = a c; M = a b c d;
(*Domain
  assumptions:the flat angles and the sigma-shifted angles lie in (0,Pi);
these hold under the hypotheses of the lemma
  indicated in the subsection above (see paper)*)
assumptions = {0 < α < Pi, 0 < β < Pi, 0 < γ < Pi,
  0 < δ < Pi, 0 < σ - α < Pi, 0 < σ - β < Pi, 0 < σ - γ < Pi, 0 < σ - δ < Pi};
rad = (f - M) (r - 1) (s - 1) / (f (1 - M));
sq = (Sin[σ] / Sin[σ - δ]) ^2;
isTrueQ[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions → assumptions]];
(*Two-stage verification,
in the order of the table:STAGE 1 (the radicand row) identifies rad with the
  perfect square sq.STAGE 2 (the identity row) uses Stage 1:for t=

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Sin[sigma]/Sin[sigma-delta], the eps-split assumptions
give t the sign eps (u/v checks),
so Sqrt[rad]=Sqrt[t^2]=eps*t (abstract root law),
and eps*Sqrt[rad]*Sin[sigma-delta]=
eps^2*Sin[sigma]=Sin[sigma] (cancellation). No Sqrt
over trigonometric arguments reaches the kernel.
isTrueQeps[lhs_, rhs_] :=
Module[{t, u, v}, TrueQ[FullSimplify[rad == sq, Assumptions -> assumptions]] &&
  TrueQ[Assuming[u > 0 && v > 0, Simplify[u/v > 0]]] &&
  TrueQ[Assuming[u < 0 && v > 0, Simplify[u/v < 0]]] &&
  TrueQ[Assuming[t > 0, Simplify[Sqrt[t^2] == t]]] &&
  TrueQ[Assuming[t < 0, Simplify[Sqrt[t^2] == -t]]] &&
  TrueQ[Simplify[(Sin[sigma]/Sin[sigma-delta]) Sin[sigma-delta] == Sin[sigma]]];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background -> RGBColor[0.13, 0.55, 0.13], FrameStyle -> None,
  RoundingRadius -> 7, FrameMargins -> {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background -> RGBColor[0.82, 0.10, 0.10], FrameStyle -> None,
  RoundingRadius -> 7, FrameMargins -> {{12, 12}, {4, 4}}];
(* =====*)
(*PART 2:THE RADICAND AND THE IDENTITY*)
(* =====*)
identities = {"Radicand equals a perfect square", HoldForm[
  (f-M) (r-1) (s-1) / (f (1-M))], HoldForm[(Sin[sigma]/Sin[sigma-delta])^2], isTrueQ},
  {"Identity for sin(sigma) (via the radicand row)", HoldForm[Sin[sigma]],
  HoldForm[Epsilon Sqrt[(f-M) (r-1) (s-1) / (f (1-M))] Sin[sigma-delta]], isTrueQeps}};
(* =====*)
(*PART 3:VERIFICATION TABLE*)
(* =====*)
row[{desc_, lhs_, rhs_, test_}] :=
  {Style[desc, 13], Item[Style[TraditionalForm[lhs], 16], Alignment -> Center],
  Item[Style[TraditionalForm[rhs], 16], Alignment -> Center],
  statusChip[test[ReleaseHold@lhs, ReleaseHold@rhs]]};
Grid[
  Join[{{Style["Description", Bold, 14], Style["Left Hand Side (LHS)", Bold, 14],
  Style["Right Hand Side (RHS)", Bold, 14],
  Style["Equality Status", Bold, 14]}}, row /@ identities],
  Frame -> All, FrameStyle -> GrayLevel[0.75],
  Background -> {None, {1 -> RGBColor[0.99, 0.96, 0.88]}},
  Spacings -> {2.2, 2.8},
  Alignment -> {{Left, Center, Center, Center}, Center},
  ItemStyle -> Directive[FontFamily -> "Helvetica", 13]]

```

Out[]=

Description	Left Hand Side (LHS)	Right Hand Side (RHS)	Equality Status
Radicand equals a perfect square	$\frac{(f-M)(r-1)(s-1)}{f(1-M)}$	$\left(\frac{\sin(\sigma)}{\sin(\sigma-\delta)}\right)^2$	Verified
Identity for $\sin(\sigma)$ (via the radicand row)	$\sin(\sigma)$	$\varepsilon \sqrt{\frac{(f-M)(r-1)(s-1)}{f(1-M)}} \sin(\sigma - \delta)$	Verified

Section 4

Results of Section 4 are used in the proof of Theorem 1 (see paper)

```
In[64]:= (* =====*)
(*PART 1:INITIALIZATION& STANDING HYPOTHESES*)
(* =====*)
ClearAll[f, i, j, l, m, c, s, si, sl, fi, fj, fl, fm,
  M, chi, x, display, propEq, standingOf, assumptions, isTrueQ,
  statusChip, caseCheck, results, starRef, fj3, fm3, fj4, fm4];
Format[fi, TraditionalForm] = Subscript[f, i];
Format[fl, TraditionalForm] = Subscript[f, l];
Format[chi, TraditionalForm] = x;
(*Standing hypotheses:M and all four f's are positive,
distinct from 1 and from each other's modulus pairing f≠M,
and each pair-product (f-1)(f-M) is positive;
these hold under the hypotheses indicated
in the paper.The verification below is branch-wise:
sign data entering through chi and the regime of M is enumerated explicitly,
so every square root reaching the kernel has a sign-determined
radicand.Parametrization:the first term of the identity carries zeta1 alone;
in every case the verifying choice is zeta1=chi (the value of chi
on the branch).All display-only rows are wrapped in HoldForm,
so they cannot evaluate through symbols defined in other sections.*)
standingOf[fjE_, fmE_] := M > 0 && M ≠ 1 && And@@ (# > 0 & /@ {fi, fjE, fl, fmE}) &&
  And@@ (# ≠ 1 & /@ {fi, fjE, fl, fmE}) && And@@ (# ≠ M & /@ {fi, fjE, fl, fmE}) &&
  And@@ ((# - 1) (# - M) > 0 & /@ {fi, fjE, fl, fmE});
isTrueQ[lhs_, rhs_] :=
  TrueQ[FullSimplify[lhs == rhs, Assumptions → assumptions]];
display[zeta1_, zeta2_, chiv_, fjE_, fmE_] :=
```

```

zeta1 M Sqrt[(fi - 1) (fjE - 1) (fl - 1) (fmE - 1)] + zeta2 (1 - M)
Sqrt[fi fjE fl fmE] - zeta2 chiv Sqrt[(fi - M) (fjE - M) (fl - M) (fmE - M)];
propEq[zeta1_, zeta2_, chiv_, fjE_, fmE_] := Style[TraditionalForm[
HoldForm[#] &@ (display[zeta1, zeta2, chiv, fjE, fmE] == M (1 - M))], 16];
starRef = Style["(*)", Bold, 16, RGBColor[0.72, 0.45, 0.02]];
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
(*Two-stage verification per
case:STAGE 1 (the collapse row) identifies the instantiated left-hand side
of (star) with its square-root-free form under the branch assumptions;
STAGE 2 (the closure row) verifies that the collapsed form equals
M(1-M).The chip is Verified only if both stages hold.*)
caseCheck[title_, col_, condRow_, zeta1_, zeta2_, branchRow_, chiv_,
fjE_, fmE_, extraAssum_, extraRow_, collapsedHeld_] := Module[
{okCollapse, okZero, ok}, assumptions = standingOf[fjE, fmE] && extraAssum;
okCollapse =
isTrueQ[display[zeta1, zeta2, chiv, fjE, fmE], ReleaseHold[collapsedHeld]];
okZero = isTrueQ[ReleaseHold[collapsedHeld], M (1 - M)];
ok = okCollapse && okZero;
Print[Panel[Column[
{Row[{Style[title, Bold, 15, col], Spacer[16], Style[condRow, 13, col]}],
Row[{Style["testing ", 13], starRef, Style[Row[{" with ",
Subscript["g", 1], " = ", TraditionalForm[zeta1], ", ",
Subscript["g", 2], " = ", TraditionalForm[zeta2]}], Bold, 13],
Spacer[24], branchRow}], propEq[zeta1, zeta2, chiv, fjE, fmE], Row[
{Style["expanding the square roots: ", GrayLevel[0.35], 12], Style[
TraditionalForm[HoldForm[#] &@ (collapsedHeld == M (1 - M))], 13}],
Row[{Style["→ ", Bold, 14], Style[TraditionalForm[HoldForm[0 == 0]],
14], Spacer[26], statusChip[ok]}],
Style[Row[{"additionally used: ", extraRow}], GrayLevel[0.35], 12}],
Spacings → 1], Background → Lighter[col, 0.93], FrameMargins → 12]];
ok];
(* =====*)
(*PART 2:STANDING HYPOTHESES AND THE IDENTITY (star)*)
(* =====*)
Print[Panel[Column[{Style["Standing Hypotheses:", Bold, 14],
Style[Row[{TraditionalForm[HoldForm[Subscript[f, i]]], ", ",
TraditionalForm[HoldForm[M]], " > 0, ", TraditionalForm[HoldForm[
Subscript[f, i]], ", ", TraditionalForm[HoldForm[M]], " ≠ 1, ",
TraditionalForm[HoldForm[Subscript[f, i] ≠ M]], ", ", TraditionalForm[
HoldForm[(Subscript[f, i] - 1) (Subscript[f, i] - M) > 0]],
" for each ", TraditionalForm[HoldForm[i]], " = 1, ..., 4, ",
TraditionalForm[HoldForm[χ ∈ {±1}]]}], 13], ""], Style[

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Row[{"The identity we want to verify for all cases below (for some ",
  Subscript["g", 1], ", ", Subscript["g", 2], " ∈ {±1}):"}, Bold, 14],
Row[{Style[TraditionalForm[HoldForm[Subscript[g, 1] M
  Sqrt[Product[Subscript[f, i] - 1, {i, 1, 4}]] + Subscript[g, 2]
  (1 - M) Sqrt[Product[Subscript[f, i], {i, 1, 4}]] - Subscript[g, 2]
  x Sqrt[Product[Subscript[f, i] - M, {i, 1, 4}]] = M (1 - M)], 16],
  Spacer[28], Style["(*)", Bold, 16, RGBColor[0.72, 0.45, 0.02]]}],
Spacings → 1.1, Background → RGBColor[0.99, 0.96, 0.88],
FrameMargins → 14]];

(* =====*)
(*PART 3:CASE-BY-CASE VERIFICATION*)
(* =====*)
results = {};
(*Condition (i):fj=fi,fm=fl;
chi (fi-1)(fl-1)>0;
zeta1=chi=si*sl per branch,
zeta2=1.The four branches enumerate the signs of fi-1 and fl-1;
by the standing pair-products,fi-M and fi-1 have the same sign,
and similarly at l,so chi=si*sl.*)
Do[AppendTo[results, caseCheck["Condition (i):", RGBColor[0.10, 0.30, 0.70],
  Row[{TraditionalForm[HoldForm[Subscript[f, j] == Subscript[f, i]]],
    ", ", TraditionalForm[HoldForm[Subscript[f, m] == Subscript[f, l]]],
    ", ", TraditionalForm[
      HoldForm[x (Subscript[f, i] - 1) (Subscript[f, l] - 1) > 0]]], si sl,
  1, Style[Row[{"branch: sign(", TraditionalForm[HoldForm[fi - 1]],
    ") = ", si, ", sign(", TraditionalForm[HoldForm[fl - 1]], ") = ", sl,
    " → x = ", si sl, ", so \!\(\*SubscriptBox[\(g\), \(\{1\}\)]\) = x"}],
  13, GrayLevel[0.3]], si sl, fi, fl, si (fi - 1) > 0 &&
  si (fi - M) > 0 && sl (fl - 1) > 0 && sl (fl - M) > 0,
  Row[{TraditionalForm[HoldForm[#] &@(si (fi - 1) > 0)], ", ",
    TraditionalForm[HoldForm[#] &@(si (fi - M) > 0)],
    ", ", TraditionalForm[HoldForm[#] &@(sl (fl - 1) > 0)],
    ", ", TraditionalForm[HoldForm[#] &@(sl (fl - M) > 0)],
    " (same-sign facts by the standing hypotheses)"}],
  HoldForm[M (fi - 1) (fl - 1) + (1 - M) fi fl - (fi - M) (fl - M)]]],
  {si, {1, -1}}, {sl, {1, -1}}];
(*Condition (ii):fi*fj=M,fl*fm=M,that is,fj=M/fi,fm=M/fl;
zeta1=chi(symbolic),zeta2=1;
no sign condition is consumed and the two chi-
dressed terms cancel syntactically.*)
AppendTo[results, caseCheck["Condition (ii):", RGBColor[0.55, 0.25, 0.70],
  Row[{TraditionalForm[HoldForm[Subscript[f, i] Subscript[f, j] == M]],
    ", ", TraditionalForm[
      HoldForm[Subscript[f, l] Subscript[f, m] == M]]], chi, 1, Style[
  Row[{"single symbolic branch, \!\(\*SubscriptBox[\(g\), \(\{1\}\)]\) = x"}],
  13, GrayLevel[0.3]], chi, M / fi, M / fl, True,
  "none (x remains symbolic; no sign condition is consumed)",

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HoldForm[chi M Sqrt[(fi - 1) (M / fi - 1) (fl - 1) (M / fl - 1)] +
  (1 - M) M - chi M Sqrt[(fi - 1) (M / fi - 1) (fl - 1) (M / fl - 1)]]];
(*Condition (iii):fi+fj-fi*fj=M,fl+fm-fl*fm=M,
that is,fj=(M-fi)/(1-fi),fm=(M-fl)/(1-fl);
chi (1-M)>0;
zeta1=chi=c per branch,zeta2=1.Two branches:chi=+1 with M<1,
and chi=-1 with M>1.*)
fj3 = (M - fi) / (1 - fi); fm3 = (M - fl) / (1 - fl);
Do[AppendTo[results, caseCheck["Condition (iii):",
  RGBColor[0.00, 0.50, 0.50], Row[{TraditionalForm[HoldForm[
    Subscript[f, i] + Subscript[f, j] - Subscript[f, i] Subscript[f, j] == M]],
    ", ", TraditionalForm[HoldForm[
    Subscript[f, l] + Subscript[f, m] - Subscript[f, l] Subscript[f, m] == M]],
    ", ", TraditionalForm[HoldForm[χ (1 - M) > 0]]]], c, 1,
Style[Row[{"branch: χ = ", c, ", ", If[c == 1, "M < 1", "M > 1"],
  ", so \!\(\*SubscriptBox[\(\xi\), \(\mathbf{1}\)]\) = χ"}],
  13, GrayLevel[0.3]], c, fj3, fm3, If[c == 1, M < 1, M > 1],
Row[{TraditionalForm[HoldForm[#] &@If[c == 1, M < 1, M > 1]],
  " (this branch of ",
  TraditionalForm[HoldForm[χ (1 - M) > 0]], ")"], HoldForm[
  M (1 - M) + (1 - M) Sqrt[fi ((M - fi) / (1 - fi)) fl ((M - fl) / (1 - fl))] - (1 - M)
  Sqrt[fi ((M - fi) / (1 - fi)) fl ((M - fl) / (1 - fl))]]], {c, {1, -1}}];
(*Condition (iv):M(fi+fj-1)=fi*fj,M(fl+fm-1)=fl*fm,
that is,fj=M(fi-1)/(fi-M),fm=M(fl-1)/(fl-M);
zeta1=chi=c per branch,and zeta2=-1 if chi(1-M)>0,
zeta2=+1 if chi(1-M)<0,encoded as zeta2=
-c*s with s=sign(1-M).Four branches over chi and the regime of M;
the collapse row uses per-branch literal templates so
the cancelling pair of square-root terms is shown.*)
fj4 = M (fi - 1) / (fi - M); fm4 = M (fl - 1) / (fl - M);
Do[AppendTo[results, caseCheck["Condition (iv):", RGBColor[0.75, 0.35, 0.05],
  Row[{TraditionalForm[HoldForm[M (Subscript[f, i] + Subscript[f, j] - 1) =
    Subscript[f, i] Subscript[f, j]]], ", ",
    TraditionalForm[HoldForm[M (Subscript[f, l] + Subscript[f, m] - 1) =
    Subscript[f, l] Subscript[f, m]]]], c, -c s,
Style[Row[{"branch: χ = ", c, ", ", If[s == 1, "M < 1", "M > 1"],
  " → ", If[c s == 1, "χ(1-M) > 0", "χ(1-M) < 0"],
  ", \!\(\*SubscriptBox[\(\xi\), \(\mathbf{1}\)]\) = χ"}],
  13, GrayLevel[0.3]], c, fj4, fm4, If[s == 1, M < 1, M > 1],
Row[{TraditionalForm[HoldForm[#] &@If[s == 1, M < 1, M > 1]],
  " (this branch of the rule: ", Subscript["ξ", 2], " = -1 if ",
  TraditionalForm[HoldForm[χ (1 - M) > 0]], ", ", Subscript["ξ", 2],
  " = +1 if ", TraditionalForm[HoldForm[χ (1 - M) < 0]], ")"], If[c s == 1,
HoldForm[(1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] -
  (1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] + M (1 - M)],
HoldForm[-(1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] +
  (1 - M) Sqrt[fi (M (fi - 1) / (fi - M)) fl (M (fl - 1) / (fl - M))] +

```

```

M (1 - M)]]]], {c, {1, -1}}, {s, {1, -1}}];
(* =====*)
(*PART 4:SUMMARY*)
(* =====*)
Print[
  Framed[Style[If[And@@results, "ALL CASES VERIFIED", "SOME CASES FAILED"],
    Bold, White, 16], Background →
    If[And@@results, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
    FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];

```

Standing Hypotheses:

$f_i, M > 0, f_i, M \neq 1, f_i \neq M, (f_i - 1)(f_i - M) > 0$ for each $i = 1, \dots, 4, \chi \in \{\pm 1\}$

The identity we want to verify for all cases below (for some $\zeta_1, \zeta_2 \in \{\pm 1\}$):

$$\zeta_1 M \sqrt{\prod_{i=1}^4 (f_i - 1)} + \zeta_2 (1 - M) \sqrt{\prod_{i=1}^4 f_i} - \zeta_2 \chi \sqrt{\prod_{i=1}^4 (f_i - M)} = M(1 - M) \quad (\star)$$

Condition (i): $f_j = f_i, f_m = f_l, \chi (f_i - 1)(f_l - 1) > 0$

testing $(\star)$ with $\zeta_1 = 1, \zeta_2 = 1$

branch: $\text{sign}(f_i - 1) = 1, \text{sign}(f_l - 1) = 1 \rightarrow \chi = 1, \text{ so } \zeta_1 = \chi$

$$\sqrt{f_i^2 f_l^2} (1 - M) - \sqrt{(f_i - M)^2 (f_l - M)^2} + \sqrt{(-1 + f_i)^2 (-1 + f_l)^2} M = (1 - M) M$$

expanding the square roots: $M(f_i - 1)(f_l - 1) + (1 - M)f_i f_l - (f_i - M)(f_l - M) = (1 - M) M$

$\rightarrow 0 = 0$

Verified

additionally used:

$-1 + f_i > 0, f_i - M > 0, -1 + f_l > 0, f_l - M > 0$ (same-sign facts by the standing hypotheses)

Condition (i): $f_j = f_i, f_m = f_l, \chi (f_i - 1)(f_l - 1) > 0$

testing $(\star)$ with $\zeta_1 = -1, \zeta_2 = 1$

branch: $\text{sign}(f_i - 1) = 1, \text{sign}(f_l - 1) = -1 \rightarrow \chi = -1, \text{ so } \zeta_1 = \chi$

$$\sqrt{f_i^2 f_l^2} (1 - M) + \sqrt{(f_i - M)^2 (f_l - M)^2} - \sqrt{(-1 + f_i)^2 (-1 + f_l)^2} M = (1 - M) M$$

expanding the square roots: $M(f_i - 1)(f_l - 1) + (1 - M)f_i f_l - (f_i - M)(f_l - M) = (1 - M) M$

$\rightarrow 0 = 0$

Verified

additionally used:

$-1 + f_i > 0, f_i - M > 0, 1 - f_l > 0, -f_l + M > 0$ (same-sign facts by the standing hypotheses)

Condition (i): $f_j = f_i, f_m = f_l, \chi(f_i - 1)(f_l - 1) > 0$

testing (★) with $\zeta_1 = -1, \zeta_2 = 1$

branch: $\text{sign}(f_i - 1) = -1, \text{sign}(f_l - 1) = 1 \rightarrow \chi = -1$, so $\zeta_1 = \chi$

$$\sqrt{f_i^2 f_l^2} (1 - M) + \sqrt{(f_i - M)^2 (f_l - M)^2} - \sqrt{(-1 + f_i)^2 (-1 + f_l)^2} M = (1 - M) M$$

expanding the square roots: $M(f_i - 1)(f_l - 1) + (1 - M)f_i f_l - (f_i - M)(f_l - M) = (1 - M) M$

$$\rightarrow 0 = 0$$

Verified

additionally used:

$1 - f_i > 0, -f_i + M > 0, -1 + f_i > 0, f_l - M > 0$ (same-sign facts by the standing hypotheses)

Condition (i): $f_j = f_i, f_m = f_l, \chi(f_i - 1)(f_l - 1) > 0$

testing (★) with $\zeta_1 = 1, \zeta_2 = 1$

branch: $\text{sign}(f_i - 1) = -1, \text{sign}(f_l - 1) = -1 \rightarrow \chi = 1$, so $\zeta_1 = \chi$

$$\sqrt{f_i^2 f_l^2} (1 - M) - \sqrt{(f_i - M)^2 (f_l - M)^2} + \sqrt{(-1 + f_i)^2 (-1 + f_l)^2} M = (1 - M) M$$

expanding the square roots: $M(f_i - 1)(f_l - 1) + (1 - M)f_i f_l - (f_i - M)(f_l - M) = (1 - M) M$

$$\rightarrow 0 = 0$$

Verified

additionally used:

$1 - f_i > 0, -f_i + M > 0, 1 - f_l > 0, -f_l + M > 0$ (same-sign facts by the standing hypotheses)

Condition (ii): $f_i f_j = M, f_l f_m = M$

testing (★) with $\zeta_1 = \chi, \zeta_2 = 1$ single symbolic branch, $\zeta_1 = \chi$

$$(1 - M) \sqrt{M^2} + \chi M \sqrt{(-1 + f_i)(-1 + f_l) \left(-1 + \frac{M}{f_i}\right) \left(-1 + \frac{M}{f_l}\right)} -$$

$$\chi \sqrt{(f_i - M)(f_l - M) \left(-M + \frac{M}{f_i}\right) \left(-M + \frac{M}{f_l}\right)} = (1 - M) M$$

expanding the square roots: $\chi M \sqrt{(f_i - 1) \left(\frac{M}{f_i} - 1\right) (f_l - 1) \left(\frac{M}{f_l} - 1\right)} +$

$$(1 - M) M - \chi M \sqrt{(f_i - 1) \left(\frac{M}{f_i} - 1\right) (f_l - 1) \left(\frac{M}{f_l} - 1\right)} = (1 - M) M$$

$$\rightarrow 0 = 0$$

Verified

additionally used: none (χ remains symbolic; no sign condition is consumed)

Condition (iii): $f_i + f_j - f_i f_j = M$, $f_l + f_m - f_l f_m = M$, $\chi(1-M) > 0$

testing (★) with $\zeta_1 = 1$, $\zeta_2 = 1$ branch: $\chi = 1$, $M < 1$, so $\zeta_1 = \chi$

$$(1-M) \sqrt{\frac{f_i f_l (-f_i+M)(-f_l+M)}{(1-f_i)(1-f_l)}} + M \sqrt{(-1+f_i)(-1+f_l)\left(-1+\frac{-f_i+M}{1-f_i}\right)\left(-1+\frac{-f_l+M}{1-f_l}\right)} - \sqrt{(f_i-M)(f_l-M)\left(-M+\frac{-f_i+M}{1-f_i}\right)\left(-M+\frac{-f_l+M}{1-f_l}\right)} = (1-M)M$$

expanding the square roots:

$$M(1-M) + (1-M) \sqrt{\frac{f_i(M-f_i)f_l(M-f_l)}{(1-f_i)(1-f_l)}} - (1-M) \sqrt{\frac{f_i(M-f_i)f_l(M-f_l)}{(1-f_i)(1-f_l)}} = (1-M)M$$

→ 0 = 0

Verified

additionally used: $M < 1$ (this branch of $\chi(1-M) > 0$)

Condition (iii): $f_i + f_j - f_i f_j = M$, $f_l + f_m - f_l f_m = M$, $\chi(1-M) > 0$

testing (★) with $\zeta_1 = -1$, $\zeta_2 = 1$ branch: $\chi = -1$, $M > 1$, so $\zeta_1 = \chi$

$$(1-M) \sqrt{\frac{f_i f_l (-f_i+M)(-f_l+M)}{(1-f_i)(1-f_l)}} - M \sqrt{(-1+f_i)(-1+f_l)\left(-1+\frac{-f_i+M}{1-f_i}\right)\left(-1+\frac{-f_l+M}{1-f_l}\right)} + \sqrt{(f_i-M)(f_l-M)\left(-M+\frac{-f_i+M}{1-f_i}\right)\left(-M+\frac{-f_l+M}{1-f_l}\right)} = (1-M)M$$

expanding the square roots:

$$M(1-M) + (1-M) \sqrt{\frac{f_i(M-f_i)f_l(M-f_l)}{(1-f_i)(1-f_l)}} - (1-M) \sqrt{\frac{f_i(M-f_i)f_l(M-f_l)}{(1-f_i)(1-f_l)}} = (1-M)M$$

→ 0 = 0

Verified

additionally used: $M > 1$ (this branch of $\chi(1-M) > 0$)

Condition (iv): $M(f_i + f_j - 1) = f_i f_j$, $M(f_l + f_m - 1) = f_l f_m$

testing (★) with $\zeta_1 = 1$, $\zeta_2 = -1$ branch: $\chi = 1$, $M < 1 \rightarrow \chi(1-M) > 0$, $\zeta_1 = \chi$

$$-\left((1-M) \sqrt{\frac{(-1+f_i)f_i(-1+f_l)f_l M^2}{(f_i-M)(f_l-M)}}\right) + M \sqrt{(-1+f_i)(-1+f_l)\left(-1+\frac{(-1+f_i)M}{f_i-M}\right)\left(-1+\frac{(-1+f_l)M}{f_l-M}\right)} + \sqrt{(f_i-M)(f_l-M)\left(-M+\frac{(-1+f_i)M}{f_i-M}\right)\left(-M+\frac{(-1+f_l)M}{f_l-M}\right)} = (1-M)M$$

expanding the square roots:

$$(1-M) \sqrt{\frac{f_i(M(f_i-1))f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} - (1-M) \sqrt{\frac{f_i(M(f_i-1))f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + M(1-M) = (1-M)M$$

→ 0 = 0

Verified

additionally used: $M < 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

Condition (iv): $M(f_i + f_j - 1) = f_i f_j$, $M(f_l + f_m - 1) = f_l f_m$

testing (★) with $\zeta_1 = 1$, $\zeta_2 = 1$ branch: $\chi = 1$, $M > 1 \rightarrow \chi(1-M) < 0$, $\zeta_1 = \chi$

$$(1-M) \sqrt{\frac{(-1+f_i) f_i (-1+f_l) f_l M^2}{(f_i-M)(f_l-M)}} +$$

$$M \sqrt{(-1+f_i)(-1+f_l) \left(-1 + \frac{(-1+f_i)M}{f_i-M}\right) \left(-1 + \frac{(-1+f_l)M}{f_l-M}\right)} -$$

$$\sqrt{(f_i-M)(f_l-M) \left(-M + \frac{(-1+f_i)M}{f_i-M}\right) \left(-M + \frac{(-1+f_l)M}{f_l-M}\right)} = (1-M)M$$

expanding the square roots:

$$-(1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + (1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + M(1-M) = (1-M)M$$

$\rightarrow 0 = 0$

Verified

additionally used: $M > 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

Condition (iv): $M(f_i + f_j - 1) = f_i f_j$, $M(f_l + f_m - 1) = f_l f_m$

testing (★) with $\zeta_1 = -1$, $\zeta_2 = 1$ branch: $\chi = -1$, $M < 1 \rightarrow \chi(1-M) < 0$, $\zeta_1 = \chi$

$$(1-M) \sqrt{\frac{(-1+f_i) f_i (-1+f_l) f_l M^2}{(f_i-M)(f_l-M)}} -$$

$$M \sqrt{(-1+f_i)(-1+f_l) \left(-1 + \frac{(-1+f_i)M}{f_i-M}\right) \left(-1 + \frac{(-1+f_l)M}{f_l-M}\right)} +$$

$$\sqrt{(f_i-M)(f_l-M) \left(-M + \frac{(-1+f_i)M}{f_i-M}\right) \left(-M + \frac{(-1+f_l)M}{f_l-M}\right)} = (1-M)M$$

expanding the square roots:

$$-(1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + (1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + M(1-M) = (1-M)M$$

$\rightarrow 0 = 0$

Verified

additionally used: $M < 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

Condition (iv): $M(f_i + f_j - 1) = f_i f_j$, $M(f_l + f_m - 1) = f_l f_m$

testing (★) with $\zeta_1 = -1$, $\zeta_2 = -1$ branch: $\chi = -1$, $M > 1 \rightarrow \chi(1-M) > 0$, $\zeta_1 = \chi$

$$\begin{aligned}
 & - \left((1-M) \sqrt{\frac{(-1+f_i) f_i (-1+f_l) f_l M^2}{(f_i-M)(f_l-M)}} \right) - \\
 & M \sqrt{(-1+f_i)(-1+f_l) \left(-1 + \frac{(-1+f_i)M}{f_i-M} \right) \left(-1 + \frac{(-1+f_l)M}{f_l-M} \right)} - \\
 & \sqrt{(f_i-M)(f_l-M) \left(-M + \frac{(-1+f_i)M}{f_i-M} \right) \left(-M + \frac{(-1+f_l)M}{f_l-M} \right)} = (1-M)M
 \end{aligned}$$

expanding the square roots:

$$(1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} - (1-M) \sqrt{\frac{f_i(M(f_i-1)) f_l(M(f_l-1))}{(f_i-M)(f_l-M)}} + M(1-M) = (1-M)M$$

$\rightarrow 0 = 0$

Verified

additionally used: $M > 1$ (this branch of the rule: $\zeta_2 = -1$ if $\chi(1-M) > 0$, $\zeta_2 = +1$ if $\chi(1-M) < 0$)

ALL CASES VERIFIED

Section 5

Results of Section 5 are used in the proof of Remark 5
(see paper)

```

In[ ]:= (* =====*)
(*PART 1:INITIALIZATION*)
(* =====*)
ClearAll[gi, gj, k, kp, ei, ej, xi, h1of, h2of, h3of, baseAssum,
  wConfigs, ePairs, dirForward, dirBackward, branchOK, smallChip,
  statusChip, panelH, resultsAll, condOf, vars, hsym, stmtOf];
Format[gi, TraditionalForm] = Subscript[g, i];
Format[gj, TraditionalForm] = Subscript[g, j];
Format[kp, TraditionalForm] = k';
statusChip[True] := Framed[Style["Verified", Bold, White, 13],
  Background → RGBColor[0.13, 0.55, 0.13], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
statusChip[False] := Framed[Style["Failed", Bold, White, 13],
  Background → RGBColor[0.82, 0.10, 0.10], FrameStyle → None,
  RoundingRadius → 7, FrameMargins → {{12, 12}, {4, 4}}];
smallChip[True] := Style["☐", 14, RGBColor[0.13, 0.55, 0.13]];
smallChip[False] := Style["x", 14, Bold, RGBColor[0.82, 0.10, 0.10]];
vars = {gi, gj, k, kp};

```

```

(*The three h's in the case  $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$ ,
with xi a numeric sign per branch.*)
h1of[e1_, e2_, x_] :=
  (e1 gj Sqrt[(gi^2 - 1) (gj^2 - kp^2) x] + e2 gi Sqrt[(gj^2 - 1) (gi^2 - kp^2) x]) /
  (k^2 - (gi^2 - 1) (gj^2 - 1));
h2of[e1_, e2_, x_] :=
  (Sqrt[(gi^2 - kp^2) (gj^2 - kp^2) x] + e1 e2 gi gj Sqrt[(gi^2 - 1) (gj^2 - 1) x]) /
  (k^2 - (gi^2 - 1) (gj^2 - 1));
h3of[e1_, e2_, x_] :=
  (k^2 gi gj + e1 e2 x Sqrt[(gi^2 - 1) (gi^2 - kp^2) (gj^2 - 1) (gj^2 - kp^2)]) /
  (k^2 - (gi^2 - 1) (gj^2 - 1));
(*Standing assumptions:k,k' with k'^2=1-k^2;
the case guard;
window assumptions added per branch.*)
baseAssum = 0 < k < 1 && kp > 0 && kp^2 == 1 - k^2 && k^2 != (gi^2 - 1) (gj^2 - 1);
wConfigs =
  {{"!\(\(*SubscriptBox[\(g\), \(\i\)]\)\), \!\(\(*SubscriptBox[\(g\), \(\j\)]\)\)
    \in (1, \infty)", gi > 1 && gj > 1, 1},
   {"!\(\(*SubscriptBox[\(g\), \(\i\)]\)\) \in (1, \infty), \!\(\(*SubscriptBox[\(g\),
    \(\j\)]\)\) \in (0, k')", gi > 1 && 0 < gj < kp, -1},
   {"!\(\(*SubscriptBox[\(g\), \(\i\)]\)\) \in (0, k'), \!\(\(*SubscriptBox[\(g\),
    \(\j\)]\)\) \in (1, \infty)", 0 < gi < kp && gj > 1, -1},
   {"!\(\(*SubscriptBox[\(g\), \(\i\)]\)\), \!\(\(*SubscriptBox[\(g\),
    \(\j\)]\)\) \in (0, k')", 0 < gi < kp && 0 < gj < kp, 1}};
ePairs = {{1, 1}, {1, -1}, {-1, 1}, {-1, -1}};
(*Two-directional branch check:no zero off the condition locus,
and no nonzero value on it.A condition False
means the claim on this branch is pure nonvanishing.*)
dirForward[h_, cond_, assum_] :=
  Reduce[assum && h == 0 && ! cond, vars, Reals] === False;
dirBackward[h_, cond_, assum_] :=
  If[cond === False, True, Reduce[assum && cond && h != 0, vars, Reals] === False];
branchOK[h_, cond_, assum_] :=
  dirForward[h, cond, assum] && dirBackward[h, cond, assum];
(*Per-branch condition of each claim;
False encodes "never vanishes".*)
condOf[1, e1_, e2_] := If[e1 == -e2, gi == gj, False];
condOf[2, e1_, e2_] := If[e1 == -e2, gi gj == kp, False];
condOf[3, e1_, e2_] :=
  If[e1 == e2, k^2 (gi^2 + gj^2) == k^2 - (gi^2 - 1) (gj^2 - 1), False];
hsym[n_] := Subscript["h", "ij" <> ToString[n]];
stmtOf[n_, e1_, e2_] := Module[{c = condOf[n, e1, e2]}, If[c === False,
  Row[{hsym[n], " != 0"}], Row[{hsym[n], " = 0 \Leftrightarrow ", TraditionalForm[c]}]]];
(* =====*)
(*PART 2:THE QUANTITIES AND THE CLAIMS TO TEST*)
(* =====*)
Print[Panel[Column[{Style[

```

```

"The quantities under test, in the case  $\backslash!(\backslash(*SuperscriptBox[\backslash(k\backslash),$ 
 $\backslash(2\backslash)]\backslash) \neq (\backslash!(\backslash(*SubsuperscriptBox[\backslash(g\backslash), \backslash(i\backslash), \backslash(2\backslash)]\backslash)$ 
 $- 1)(\backslash!(\backslash(*SubsuperscriptBox[\backslash(g\backslash), \backslash(j\backslash), \backslash(2\backslash)]\backslash) - 1):$ ",
Bold, 14], Style[TraditionalForm[HoldForm[Subscript[h, ij1] ==
(Subscript[e, i]  $\times$  Subscript[g, j] Sqrt[(Subscript[g, i]^2 - 1)
(Subscript[g, j]^2 - k'^2) Subscript[ $\xi$ , ij]] +
Subscript[e, j]  $\times$  Subscript[g, i] Sqrt[(Subscript[g, j]^2 - 1)
(Subscript[g, i]^2 - k'^2) Subscript[ $\xi$ , ij]]] /
(k^2 - (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1))], 14],
Style[TraditionalForm[HoldForm[Subscript[h, ij2] ==
(Sqrt[(Subscript[g, i]^2 - k'^2) (Subscript[g, j]^2 - k'^2)
Subscript[ $\xi$ , ij]] + Subscript[e, i]  $\times$ 
Subscript[e, j]  $\times$  Subscript[g, i]  $\times$  Subscript[g, j] Sqrt[
(Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1) Subscript[ $\xi$ , ij]]] /
(k^2 - (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1))], 14],
Style[TraditionalForm[HoldForm[Subscript[h, ij3] ==
(k^2 Subscript[g, i]  $\times$  Subscript[g, j] +
Subscript[e, i]  $\times$  Subscript[e, j]  $\times$  Subscript[ $\xi$ , ij]
Sqrt[(Subscript[g, i]^2 - 1) (Subscript[g, i]^2 - k'^2)
(Subscript[g, j]^2 - 1) (Subscript[g, j]^2 - k'^2)]] /
(k^2 - (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1))], 14],
Style["and, in the case  $\backslash!(\backslash(*SuperscriptBox[\backslash(k\backslash), \backslash(2\backslash)]\backslash)$ 
 $= (\backslash!(\backslash(*SubsuperscriptBox[\backslash(g\backslash), \backslash(i\backslash), \backslash(2\backslash)]\backslash)$ 
 $- 1)(\backslash!(\backslash(*SubsuperscriptBox[\backslash(g\backslash), \backslash(j\backslash), \backslash(2\backslash)]\backslash)$ 
 $- 1)$  with  $\backslash!(\backslash(*SubscriptBox[\backslash(e\backslash), \backslash(i\backslash)]\backslash) =$ 
 $-\backslash!(\backslash(*SubscriptBox[\backslash(e\backslash), \backslash(j\backslash)]\backslash):$ ", Bold, 14],
Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij1] ==
Subscript[e, i] (Subscript[g, i]^2 - Subscript[g, j]^2) /
(2 k Subscript[g, i]  $\times$  Subscript[g, j])],
" ", TraditionalForm[HoldForm[Subscript[h, ij2] ==
(Subscript[g, i]^2  $\times$  Subscript[g, j]^2 - k'^2) /
(2 k Subscript[g, i]  $\times$  Subscript[g, j])], " ", TraditionalForm[
HoldForm[Subscript[h, ij3] == (Subscript[g, i]^2 + Subscript[g, j]^2) /
(2 Subscript[g, i]  $\times$  Subscript[g, j])]], 14],
"", Style["Standing Hypotheses:", Bold, 14], Style[
Row[{"k  $\in$  (0, 1), ", TraditionalForm[HoldForm[k'^2 == 1 - k^2]],
", ", TraditionalForm[HoldForm[Subscript[g, i]]],
", ", TraditionalForm[HoldForm[Subscript[g, j]]],
"  $\in$  (0, k')  $\cup$  (1,  $\infty$ ), ",
TraditionalForm[HoldForm[Subscript[e, i]]],
", ", TraditionalForm[HoldForm[Subscript[e, j]]],
"  $\in$  { $\pm 1$ }, ", TraditionalForm[HoldForm[Subscript[ $\xi$ , ij] ==
Sign[(Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1)]]], 13], "",
Style["The claims we test (over all window configurations of
 $\backslash!(\backslash(*SubscriptBox[\backslash(g\backslash), \backslash(i\backslash)]\backslash), \backslash!(\backslash(*SubscriptBox[\backslash(g\backslash),$ 
 $\backslash(j\backslash)]\backslash)$  and all  $\backslash!(\backslash(*SubscriptBox[\backslash(e\backslash), \backslash(i\backslash)]\backslash),$ 
 $\backslash!(\backslash(*SubscriptBox[\backslash(e\backslash), \backslash(j\backslash)]\backslash)$ , in both directions):",

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    Bold, 14], Style[TraditionalForm[HoldForm[Subscript[h, ij1] == 0 ⇔
      Subscript[g, i] == Subscript[g, j] &&
      Subscript[e, i] == -Subscript[e, j]]], 15],
    Style[TraditionalForm[HoldForm[Subscript[h, ij2] == 0 ⇔
      Subscript[g, i] × Subscript[g, j] == k' &&
      Subscript[e, i] == -Subscript[e, j]]], 15],
    Style[TraditionalForm[HoldForm[Subscript[h, ij3] == 0 ⇔
      k^2 (Subscript[g, i]^2 + Subscript[g, j]^2) ==
      k^2 - (Subscript[g, i]^2 - 1) (Subscript[g, j]^2 - 1) &&
      Subscript[e, i] == Subscript[e, j]]], 15]],
    Spacings → 1.1], Background → RGBColor[0.99,
    0.96,
    0.88], FrameMargins → 14]]];

(* =====*)
(*PART 3:CASE k^2≠(gi^2-1)(gj^2-1)*)
(* =====*)
resultsAll = {};
panelH[n_, hfun_, col_, title_] := Module[{grid, ok, headRow, stmtRow},
  grid = Table[branchOK[hfun[ep[[1]], ep[[2]], w[[3]], condOf[n, ep[[1]], ep[[2]],
    baseAssum&&w[[2]], {w, wConfigs}, {ep, ePairs}]],
  ok = And@@Flatten[grid];
  AppendTo[resultsAll, ok];
  headRow = Prepend[
    Table[Style[Row[{"(", ep[[1]], " ", "(", ep[[2]], ")"}], 12, Bold], {ep, ePairs}],
    Style["(!\\(\\*SubscriptBox[(e\\), \\(i\\)]\\), \\!\\(\\*SubscriptBox[(e\\),
      \\(j\\)]\\)) →", 12, GrayLevel[0.3]]];
  stmtRow = Prepend[Table[Style[stmtOf[n, ep[[1]], ep[[2]], 11, GrayLevel[0.25]],
    {ep, ePairs}], Style["tested →", 11, GrayLevel[0.3]]];
  Print[Panel[
    Column[{Row[{Style[title, Bold, 15, col], Spacer[20], statusChip[ok]}],
      Grid[Join[{headRow}, {stmtRow}], Table[Prepend[smallChip/@grid[[iw]],
        Style[wConfigs[[iw, 1]], 12, GrayLevel[0.3]]], {iw, 4}]],
      Alignment → Left, Spacings → {2, 0.7}]], Spacings → 1],
    Background → Lighter[col, 0.93], FrameMargins → 12]]];
panelH[1, h1of, RGBColor[0.10, 0.30, 0.70],
  "First claim (case \\(\\*SuperscriptBox[(k\\),
    \\(2\\)]\\) ≠ \\(\\*SubsuperscriptBox[(g\\), \\(i\\),
    \\(2\\)]\\)-1) \\(\\*SubsuperscriptBox[(g\\), \\(j\\), \\(2\\)]\\)-1):"];
panelH[2, h2of, RGBColor[0.55, 0.25, 0.70],
  "Second claim (case \\(\\*SuperscriptBox[(k\\),
    \\(2\\)]\\) ≠ \\(\\*SubsuperscriptBox[(g\\), \\(i\\),
    \\(2\\)]\\)-1) \\(\\*SubsuperscriptBox[(g\\), \\(j\\), \\(2\\)]\\)-1):"];
panelH[3, h3of, RGBColor[0.00, 0.50, 0.50],
  "Third claim (case \\(\\*SuperscriptBox[(k\\),
    \\(2\\)]\\) ≠ \\(\\*SubsuperscriptBox[(g\\), \\(i\\),
    \\(2\\)]\\)-1) \\(\\*SubsuperscriptBox[(g\\), \\(j\\), \\(2\\)]\\)-1):"];
(* =====*)

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(*PART 4:CASE k^2==(gi^2-1)(gj^2-1) WITH ei=-ej*)
(* =====*)
Module[{assum7, ok1, ok2, ok3, ok4, ok},
  assum7 = 0 < k < 1 && kp > 0 && kp^2 == 1 - k^2 && k^2 == (gi^2 - 1)(gj^2 - 1) &&
    ((gi > 1 && gj > 1) || (0 < gi < kp && 0 < gj < kp));
  ok1 = And @@ Table[
    branchOK[e1 (gi^2 - gj^2) / (2 k gi gj), gi == gj, assum7], {e1, {1, -1}}];
  ok2 = branchOK[(gi^2 gj^2 - kp^2) / (2 k gi gj), gi gj == kp, assum7];
  ok3 = Reduce[assum7 && (gi^2 + gj^2) / (2 gi gj) == 0, vars, Reals] == False;
  ok4 = Reduce[assum7 && k^2 (gi^2 + gj^2) == k^2 - (gi^2 - 1)(gj^2 - 1),
    vars, Reals] == False;
  ok = ok1 && ok2 && ok3 && ok4;
  AppendTo[resultsAll, ok];
  Print[
    Panel[Column[{Row[{Style["Case \!\(\*SuperscriptBox[\(k\), \((2\)]\) =
      (\!\(\*SubsuperscriptBox[\(g\), \((i\),
        \((2\)]\) - 1) (\!\(\*SubsuperscriptBox[\(g\), \((j\),
        \((2\)]\) - 1), with \!\(\*SubscriptBox[\(e\), \((i\)]\)
        = -\!\(\*SubscriptBox[\(e\), \((j\)]\)]:", Bold, 15,
        RGBColor[0.75, 0.35, 0.05]], Spacer[20], statusChip[ok]}],
      Column[{Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij1] ==
        Subscript[e, i] (Subscript[g, i]^2 - Subscript[g, j]^2) /
        (2 k Subscript[g, i] × Subscript[g, j])]],
        ": ", hsym[1], " = 0 ⇔ ", TraditionalForm[HoldForm[
        Subscript[g, i] == Subscript[g, j]]], " ", smallChip[ok1]}], 13],
        Style[Row[{TraditionalForm[HoldForm[Subscript[h, ij2] ==
        (Subscript[g, i]^2 × Subscript[g, j]^2 - k'^2) / (2 k Subscript[
        g, i] × Subscript[g, j])]], ": ", hsym[2], " = 0 ⇔ ",
        TraditionalForm[HoldForm[Subscript[g, i] × Subscript[g, j] == k']],
        " ", smallChip[ok2]}], 13], Style[
        Row[{TraditionalForm[HoldForm[Subscript[h, ij3] == (Subscript[g, i]^2 +
        Subscript[g, j]^2) / (2 Subscript[g, i] × Subscript[g, j])]],
        " ≥ 1, so ", hsym[3], " ≠ 0 ", smallChip[ok3]}], 13],
        Style[Row[{"condition of the third claim unsatisfiable here ",
        smallChip[ok4]}], 13]}], Spacings → 0.8}},
    Spacings → 1], Background → Lighter[RGBColor[0.75, 0.35,
    0.05],
    0.93], FrameMargins → 12]]];];
(* =====*)
(*PART 5:SUMMARY*)
(* =====*)
Print[Framed[Style[If[And @@ resultsAll, "ALL CLAIMS VERIFIED",
  "SOME CLAIMS FAILED"], Bold, White, 16], Background →
  If[And @@ resultsAll, RGBColor[0.13, 0.55, 0.13], RGBColor[0.82, 0.10, 0.10]],
  FrameStyle → None, RoundingRadius → 10, FrameMargins → {{18, 18}, {8, 8}}]];

```

The quantities under test, in the case $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$:

$$h_{ij1} = \frac{e_i g_j \sqrt{(g_i^2 - 1)(g_j^2 - (k')^2)} \xi_{ij} + e_j g_i \sqrt{(g_i^2 - 1)(g_j^2 - (k')^2)} \xi_{ij}}{k^2 - (g_i^2 - 1)(g_j^2 - 1)}$$

$$h_{ij2} = \frac{\sqrt{(g_i^2 - (k')^2)(g_j^2 - (k')^2)} \xi_{ij} + e_i e_j g_i g_j \sqrt{(g_i^2 - 1)(g_j^2 - 1)} \xi_{ij}}{k^2 - (g_i^2 - 1)(g_j^2 - 1)}$$

$$h_{ij3} = \frac{k^2 g_i g_j + e_i e_j \xi_{ij} \sqrt{(g_i^2 - 1)(g_j^2 - (k')^2)(g_j^2 - 1)(g_i^2 - (k')^2)}}{k^2 - (g_i^2 - 1)(g_j^2 - 1)}$$

and, in the case $k^2 = (g_i^2 - 1)(g_j^2 - 1)$ with $e_i = -e_j$:

$$h_{ij1} = \frac{e_i (g_i^2 - g_j^2)}{2 k g_i g_j} \quad h_{ij2} = \frac{g_i^2 g_j^2 - (k')^2}{2 k g_i g_j} \quad h_{ij3} = \frac{g_i^2 + g_j^2}{2 g_i g_j}$$

Standing Hypotheses:

$$k \in (0, 1), \quad (k')^2 = 1 - k^2, \quad g_i, g_j$$

$$\in (0, k') \cup (1, \infty), \quad e_i, e_j \in \{\pm 1\}, \quad \xi_{ij} = \text{sgn}((g_i^2 - 1)(g_j^2 - 1))$$

The claims we test (over all window

configurations of g_i, g_j and all e_i, e_j , in both directions):

$$h_{ij1} = 0 \Leftrightarrow g_i = g_j \wedge e_i = -e_j$$

$$h_{ij2} = 0 \Leftrightarrow g_i g_j = k' \wedge e_i = -e_j$$

$$h_{ij3} = 0 \Leftrightarrow k^2 (g_i^2 + g_j^2) = k^2 - (g_i^2 - 1)(g_j^2 - 1) \wedge e_i = e_j$$

First claim (case $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$):

Verified

$(e_i, e_j) \rightarrow$	(1, 1)	(1, -1)	(-1, 1)	(-1, -1)
tested $\rightarrow$	$h_{ij1} \neq 0$	$h_{ij1} = 0 \Leftrightarrow g_i = g_j$	$h_{ij1} = 0 \Leftrightarrow g_i = g_j$	$h_{ij1} \neq 0$
$g_i, g_j \in (1, \infty)$	✓	✓	✓	✓
$g_i \in (1, \infty), g_j \in (0, k')$	✓	✓	✓	✓
$g_i \in (0, k'), g_j \in (1, \infty)$	✓	✓	✓	✓
$g_i, g_j \in (0, k')$	✓	✓	✓	✓

Second claim (case $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$):

Verified

$(e_i, e_j) \rightarrow$	(1, 1)	(1, -1)	(-1, 1)	(-1, -1)
tested $\rightarrow$	$h_{ij2} \neq 0$	$h_{ij2} = 0 \Leftrightarrow g_i g_j = k'$	$h_{ij2} = 0 \Leftrightarrow g_i g_j = k'$	$h_{ij2} \neq 0$
$g_i, g_j \in (1, \infty)$	✓	✓	✓	✓
$g_i \in (1, \infty), g_j \in (0, k')$	✓	✓	✓	✓
$g_i \in (0, k'), g_j \in (1, \infty)$	✓	✓	✓	✓
$g_i, g_j \in (0, k')$	✓	✓	✓	✓

Third claim (case $k^2 \neq (g_i^2 - 1)(g_j^2 - 1)$):

Verified

$(e_i, e_j) \rightarrow$	(1, 1)	(1, -1)	(-1, 1)	(-1, -1)
tested $\rightarrow$	$h_{ij3} = 0 \Leftrightarrow k^2 (g_i^2 + g_j^2) = k^2 - (g_i^2 - 1)(g_j^2 - 1)$	$h_{ij3} \neq 0$	$h_{ij3} \neq 0$	$h_{ij3} = 0 \Leftrightarrow k^2 (g_i^2 + g_j^2) = k^2 - (g_i^2 - 1)(g_j^2 - 1)$
$g_i, g_j \in (1, \infty)$	✓	✓	✓	✓
$g_i \in (1, \infty), g_j \in (0, k')$	✓	✓	✓	✓
$g_i \in (0, k'), g_j \in (1, \infty)$	✓	✓	✓	✓
$g_i, g_j \in (0, k')$	✓	✓	✓	✓

Case $k^2 = (g_i^2 - 1)(g_j^2 - 1)$, with $e_i = -e_j$:

Verified

$$h_{ij1} = \frac{e_i(g_i^2 - g_j^2)}{2k g_i g_j}: h_{ij1} = 0 \Leftrightarrow g_i = g_j \quad \checkmark$$

$$h_{ij2} = \frac{g_i^2 g_j^2 - (k')^2}{2k g_i g_j}: h_{ij2} = 0 \Leftrightarrow g_i g_j = k' \quad \checkmark$$

$$h_{ij3} = \frac{g_i^2 + g_j^2}{2g_i g_j} \geq 1, \text{ so } h_{ij3} \neq 0 \quad \checkmark$$

condition of the third claim unsatisfiable here $\checkmark$

ALL CLAIMS VERIFIED